

ASSIGNMENT-8

Task – 1: Cloud Computing Research

SOLUTION-

Introduction

Cloud Computing is a technology that provides computing resources such as servers, storage, databases, networking, software, and applications over the Internet. Instead of purchasing and maintaining physical hardware, users can access these resources on demand and pay only for what they use. Cloud computing helps organizations reduce costs, improve scalability, and increase flexibility.

According to the **National Institute of Standards and Technology (NIST)**, cloud computing is a model that enables convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort.

Five Essential Characteristics of Cloud Computing (NIST)

1. On-Demand Self-Service

Users can provision computing resources such as storage, virtual machines, and applications automatically whenever required without human interaction with the service provider.

Example:

Creating an AWS EC2 instance within a few minutes.

2. Broad Network Access

Cloud services are accessible over the Internet through various devices such as laptops, mobile phones, tablets, and desktops.

Example:

Accessing Google Drive from a smartphone or laptop.

3. Resource Pooling

Cloud providers use a shared pool of resources to serve multiple customers. Resources are dynamically assigned based on user demand.

Example:

Multiple users sharing storage resources in Microsoft Azure.

4. Rapid Elasticity

Resources can be increased or decreased quickly according to workload requirements.

Example:

An e-commerce website automatically increasing server capacity during a sale.

5. Measured Service

Cloud systems automatically monitor and measure resource usage, allowing customers to pay only for what they consume.

Example:

AWS billing based on CPU, storage, and network usage.

Cloud Deployment Models

Deployment models describe how cloud infrastructure is deployed and managed.

1. Public Cloud

A Public Cloud is owned and managed by a third-party cloud provider. Resources are shared among multiple customers and are accessible over the Internet.

Features

- Low cost
- Easy scalability
- No infrastructure management

Real-World Example

[Amazon Web Services \(AWS\)](#)

2. Private Cloud

A Private Cloud is dedicated to a single organization. The infrastructure is used exclusively by one company and provides greater control and security.

Features

- High security
- Better control
- Customized environment

Real-World Example

[OpenStack](#) deployed within a company data center.

3. Hybrid Cloud

A Hybrid Cloud combines Public and Private Clouds, allowing data and applications to move between them.

Features

- Greater flexibility
- Improved scalability
- Enhanced security

Real-World Example

A company stores sensitive data in a private cloud while using AWS for additional computing power during peak demand.

Cloud Service Models

Cloud service models define the level of control and responsibility shared between the cloud provider and the customer.

1. Infrastructure as a Service (IaaS)

IaaS provides virtualized computing resources such as servers, storage, and networking over the Internet. Users manage operating systems and applications, while the provider manages the infrastructure.

Features

- Virtual machines
- Storage services
- Networking services

Examples

- [Amazon EC2](#)
- [Microsoft Azure Virtual Machines](#)

2. Platform as a Service (PaaS)

PaaS provides a complete development and deployment platform. Developers can focus on coding without worrying about hardware or operating system management.

Features

- Application development platform
- Database integration
- Automated deployment

Examples

- [Google App Engine](#)
- [Heroku](#)

3. Software as a Service (SaaS)

SaaS delivers software applications over the Internet. Users access the software through a web browser without installing or maintaining it.

Features

- Accessible from anywhere
- No installation required
- Subscription-based model

Examples

- [Google Workspace](#)
- [Microsoft 365](#)

Task – 2: AWS Account Setup & Console Exploration

SOLUTION-

Introduction

Amazon Web Services (AWS) is the world's leading cloud computing platform that provides on-demand computing resources such as servers, storage, databases, networking, security, and application services. AWS offers a **Free Tier** that allows users to explore cloud services with limited usage at no cost. To improve security and cost management, users should enable **Multi-Factor Authentication (MFA)** and configure **billing alarms**.

AWS Account Setup

1. Create an AWS Free Tier Account

An AWS Free Tier account was created using an email address and payment method. The account provides access to AWS services with limited free usage for learning and practice purposes.

Benefits

- Access to AWS Management Console
- Free usage of selected services
- Hands-on cloud learning environment

2. Enable MFA (Multi-Factor Authentication)

MFA adds an extra layer of security by requiring both a password and a verification code generated by an authentication application.

Benefits

- Protects the AWS root account
- Prevents unauthorized access
- Improves account security

3. Configure Billing Alarm (\$5)

A billing alarm was created using Amazon CloudWatch to monitor AWS charges.

Configuration

Threshold: \$5

Service: CloudWatch Billing Alarm

Notification: Email Alert

Benefits

- Tracks AWS spending
- Prevents unexpected charges
- Sends alerts when costs exceed the threshold

AWS Services Exploration

1. EC2 (Elastic Compute Cloud)

EC2 provides virtual servers in the cloud that allow users to run applications and operating systems without purchasing physical hardware.

It offers scalable computing capacity and is commonly used for hosting websites, applications, and development environments.

2. S3 (Simple Storage Service)

S3 is an object storage service used to store and retrieve data from anywhere over the Internet.

It provides high durability, scalability, and security for backups, media files, and application data.

3. RDS (Relational Database Service)

RDS is a managed database service that simplifies database administration tasks such as backups, patching, and monitoring.

It supports popular database engines such as MySQL, PostgreSQL, MariaDB, Oracle, and SQL Server.

4. IAM (Identity and Access Management)

IAM is a security service that controls access to AWS resources through users, groups, roles, and policies.

It helps administrators manage permissions and implement the principle of least privilege.

5. VPC (Virtual Private Cloud)

VPC allows users to create an isolated virtual network within AWS.

It provides complete control over IP addresses, subnets, routing tables, and network security settings.

6. CloudWatch

CloudWatch is a monitoring and observability service that collects metrics, logs, and events from AWS resources.

It helps administrators monitor system performance and create alarms for specific conditions.

7. ELB (Elastic Load Balancer)

ELB automatically distributes incoming application traffic across multiple servers or instances.

It improves application availability, scalability, and fault tolerance.

8. Lambda

AWS Lambda is a serverless computing service that executes code in response to events without requiring server management.

Users pay only for the execution time consumed by their functions.

9. Route 53

Route 53 is AWS's scalable Domain Name System (DNS) service.

It routes user requests to applications and manages domain registration and DNS records.

10. CloudFormation

CloudFormation is an Infrastructure as Code (IaC) service that allows users to define and deploy AWS resources using templates.

It automates infrastructure provisioning and ensures consistent environment deployment.

Summary Table

Service	Description
EC2	Provides virtual servers for running applications in the cloud.
S3	Object storage service for storing and retrieving files.
RDS	Managed relational database service.
IAM	Manages users, roles, and permissions in AWS.
VPC	Creates a private virtual network within AWS.
CloudWatch	Monitors AWS resources and applications.
ELB	Distributes traffic across multiple servers.

Service	Description
Lambda	Runs code without managing servers.
Route 53	Provides DNS and domain management services.
CloudFormation	Automates infrastructure deployment using templates.

Task – 3: Virtualization vs Containers Comparison-

SOLUTION-

Introduction

Virtualization and Containerization are two important technologies used in cloud computing and DevOps. Both technologies help run applications efficiently, but they work differently. Virtualization creates complete virtual machines with their own operating systems, while containers share the host operating system and package only the application and its dependencies. These technologies improve resource utilization, scalability, and deployment speed.

What is Virtualization?

Virtualization is a technology that allows multiple virtual machines (VMs) to run on a single physical machine using a software layer called a **Hypervisor**. Each virtual machine has its own operating system, libraries, and applications.

Features of Virtual Machines

- Each VM contains a complete operating system.
- Strong isolation between applications.
- Can run different operating systems on the same hardware.
- Requires more CPU, RAM, and storage resources.
- Startup time is relatively slow.

Examples

- VMware
- VirtualBox

- Microsoft Hyper-V
- AWS EC2

Typical Use Cases

- Cloud Servers
- Testing multiple operating systems
- Enterprise applications
- Legacy software support

What are Containers?

Containers are lightweight software packages that contain an application and all its dependencies. Unlike virtual machines, containers share the host operating system kernel instead of installing a separate operating system.

Features of Containers

- Lightweight architecture.
- Fast startup and deployment.
- Efficient resource utilization.
- Easy portability across environments.
- Suitable for DevOps and Microservices.

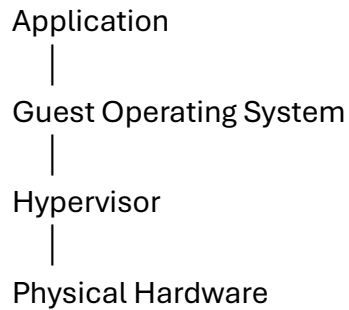
Examples

- Docker
- Kubernetes
- Podman

Typical Use Cases

- CI/CD Pipelines
- Microservices Architecture
- Application Deployment
- Cloud-Native Applications

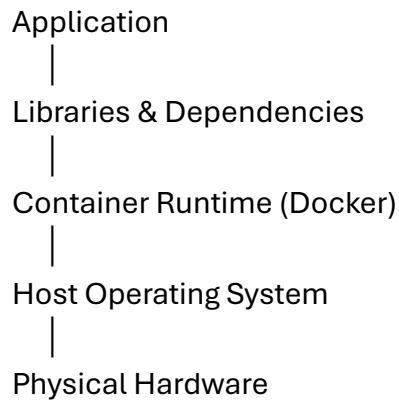
Architecture of Virtual Machines



Explanation

The Hypervisor sits between the hardware and virtual machines. Each virtual machine contains its own operating system and behaves like an independent computer.

Architecture of Containers



Explanation

Containers share the host operating system kernel, making them lightweight and faster than virtual machines.

Advantages of Virtual Machines

1. Strong security and isolation.
2. Supports multiple operating systems.
3. Suitable for traditional applications.
4. Better for complete system virtualization.

Disadvantages

1. Consumes more resources.
2. Requires large storage space.
3. Slow startup time.

Advantages of Containers

1. Lightweight and fast.
2. Uses fewer resources.
3. Easy deployment and scaling.
4. Highly portable.
5. Ideal for DevOps and cloud-native applications.

Disadvantages

1. Less isolation compared to VMs.
2. Shares the host operating system kernel.
3. Cannot run a different kernel than the host OS.

AWS Hypervisor Technology and EC2

Amazon EC2 (Elastic Compute Cloud) uses **Hypervisor Technology** to create virtual machines in the AWS Cloud.

A **Hypervisor** is software that allows multiple virtual machines to run on a single physical server. AWS uses virtualization technology to divide physical hardware into multiple EC2 instances. Each EC2 instance receives virtual CPU, memory, storage, and networking resources.

Earlier AWS used the **Xen Hypervisor**, while modern AWS infrastructure uses the **AWS Nitro System**, which improves performance, security, and resource management.

How AWS Uses Hypervisors

1. AWS data centers contain powerful physical servers.

2. Hypervisors create multiple virtual machines on these servers.
3. Each EC2 instance operates independently.
4. Users can launch, stop, restart, or terminate instances whenever required.
5. Multiple customers securely share the same physical infrastructure.

Benefits of Hypervisor Technology in AWS

- Better resource utilization.
- Strong isolation between customers.
- High scalability.
- Improved security.
- Cost-effective cloud infrastructure.

Task – 4: AWS IAM Basics-

SOLUTION-

Introduction

IAM (Identity and Access Management) is an AWS service used to securely manage access to AWS resources. IAM allows administrators to create users, groups, roles, and policies, ensuring that users receive only the permissions they need. Following the principle of **least privilege** improves security and reduces the risk of unauthorized access.

What is IAM?

AWS IAM enables organizations to:

- Create and manage AWS users.
- Create groups for easier permission management.
- Assign policies to users and groups.
- Control access to AWS resources.
- Improve cloud security through authentication and authorization.

Step 1: Create a New IAM User

Navigate to:

AWS Console → IAM → Users → Create User

User Details:

User Name : dev-user

The user was created successfully and can now access AWS based on assigned permissions.

Purpose

Creating separate IAM users instead of using the root account improves security and accountability.

Step 2: Create a Group

Navigate to:

IAM → User Groups → Create Group

Group Details:

Group Name : developers

The group was created successfully.

Purpose

Groups simplify permission management by allowing policies to be assigned to multiple users at once.

Step 3: Add User to Group

After creating the group, the user **dev-user** was added to the **developers** group.

User : dev-user

Group : developers

Purpose

The user automatically inherits all permissions assigned to the group.

Step 4: Attach ReadOnlyAccess Policy

Navigate to:

IAM → User Groups → developers → Add Permissions

Attach AWS Managed Policy:

ReadOnlyAccess

Purpose

The ReadOnlyAccess policy allows users to:

- View AWS resources.
- Monitor services.
- Read configurations.

The user cannot:

- Create resources.
- Modify resources.
- Delete resources.

This policy is useful for auditors, trainees, and monitoring teams.

Step 5: Download IAM Credential Report

Navigate to:

IAM → Credential Report

Click:

Generate Report

Then:

Download Report

The credential report contains security-related information for all IAM users.

IAM Credential Report Information

The report typically contains:

- User Name
- Password Enabled
- Password Last Used
- MFA Status
- Access Key Status
- Access Key Last Used
- Account Creation Date
- Password Rotation Status

Security Issues Identified

While reviewing the IAM Credential Report, the following common security issues may be observed:

1. MFA Not Enabled

MFA Active = FALSE

Risk

Accounts without Multi-Factor Authentication are more vulnerable to unauthorized access.

Recommendation

Enable MFA for all users and the root account.

2. Unused Access Keys

Access Key Active = TRUE

Last Used = N/A

Risk

Unused access keys can become security risks if exposed.

Recommendation

Delete unused access keys.

3. Password Not Rotated

Password Rotation Required = FALSE

Risk

Old passwords increase the likelihood of compromise.

Recommendation

Implement regular password rotation policies.

4. Root Account Without MFA

Root MFA = Disabled

Risk

The root account has full administrative privileges.

Recommendation

Always enable MFA on the root account.

5. Inactive Users

Last Login = Never

Risk

Unused accounts may become attack targets.

Recommendation

Remove or disable inactive accounts.

Benefits of Using IAM Groups

- Easier permission management.
- Reduced administrative effort.

- Consistent access control.
- Improved security.
- Supports the principle of least privilege.

Summary

Component	Description
IAM User	Individual AWS account identity.
dev-user	Created IAM user for this task.
IAM Group	Collection of users with shared permissions.
developers	Group created for development users.
ReadOnlyAccess	AWS managed policy allowing read-only access.
Credential Report	Security report containing IAM user information.
MFA	Additional security layer using verification codes.

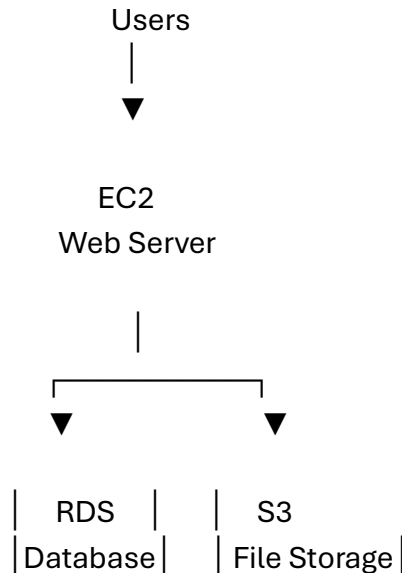
Task – 5: Cloud Architecture Documentation

SOLUTION-

Introduction

A cloud architecture defines how different cloud services work together to deliver an application. For a basic web application, AWS services such as **EC2**, **RDS**, and **S3** can be used to provide computing power, database storage, and file storage. This architecture is scalable, reliable, and suitable for small business websites or learning projects.

Cloud Architecture Diagram



Components

- **Users** access the application through a web browser.
- **EC2** hosts the website or application.
- **RDS** stores application data.
- **S3** stores images, videos, backups, and static files.

Service Justification

1. Amazon EC2 (Web Server)

Amazon EC2 provides virtual servers that host the web application. It allows users to install software, configure operating systems, and manage applications according to business requirements.

EC2 is chosen because it is scalable, reliable, and available under the AWS Free Tier, making it ideal for hosting small web applications and learning environments.

2. Amazon RDS (Database)

Amazon RDS is a managed database service that stores application data such as user accounts, product details, and transaction records. AWS automatically handles backups, updates, and database maintenance.

RDS is chosen because it reduces administrative effort and provides secure, highly available database services without requiring manual management.

3. Amazon S3 (File Storage)

Amazon S3 is an object storage service used for storing images, videos, documents, backups, and application files. Data stored in S3 is highly durable and accessible from anywhere.

S3 is chosen because it offers virtually unlimited storage, high durability, and low cost while simplifying file management.

Application Workflow

1. A user sends a request to the website.
2. The request reaches the EC2 instance hosting the application.
3. The application retrieves or stores structured data in RDS.
4. Images, documents, and static content are stored in S3.
5. The response is sent back to the user.

Estimated AWS Free Tier Usage

Service	Free Tier Limit	Estimated Monthly Usage
EC2	750 hours/month	720 hours/month
RDS	750 hours/month	720 hours/month
S3	5 GB Storage	2–3 GB Storage
Data Transfer	15 GB Outbound	5–10 GB

Estimated Cost

EC2 : Free Tier

RDS : Free Tier

S3 : Free Tier

Total : \$0 (Within Free Tier Limits)

Advantages of This Architecture

- Simple and easy to deploy.
- Fully covered by AWS Free Tier for learning purposes.
- Scalable according to user demand.
- Secure data storage.
- High availability and reliability.
- Suitable for small business websites and student projects.